Huawei Access Node BAS-ACC-37 Optical Transport Failure SOP

Node: BAS-ACC-37 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_6883

Description

The Huawei Access Node BAS-ACC-37 at the Basavanagudi POP may experience Optical Transport Failure due to degradation of SFP/XFP modules or fiber cuts resulting in loss of Rx/Tx power, leading to alarm ALARM_6883 of severity Warning. This failure occurs because of the protocol and hardware limitations inherent in the Huawei NE40E Access Node, specifically the NE40E's reliance on OTN (Optical Transport Network) technology for optical transport and data transmission. The NE40E's OTN technology employs SFP/XFP modules for converting electrical signals into optical signals and transmitting them over fiber optic cables. These modules are susceptible to degradation over time, leading to a decrease in Rx/Tx power, which in turn causes the NE40E to trigger the ALARM_6883 alarm.

Pre-requisite Checks

- Verify that the Basavanagudi POP is accessible and the BAS-ACC-37 node is reachable via SSH
- Check the status of redundant links and ensure that they are functioning properly
- Confirm that the SFP/XFP modules are securely seated and free from any physical damage

Potential Root Causes

Cause	Probability
Degradation of SFP/XFP modules	High
Fiber cuts	Medium
Rx/Tx power loss due to hardware or software issues	Low

Diagnostic Commands & Expected Outputs

```
Command: display optical-transport interface, Expected Output: Interface Name Status Tx Power (dBm) Rx Power (dBm), Description: This command displays the status and power levels of the optical transport interfaces, allowing the technician to identify any issues with the SFP/XFP modules or fiber connections.

Command: display optical-transport interface , Expected Output: Interface Name: Status: Down Tx Power (dBm): 0.00 Rx Power (dBm): 0.00, Description: This command displays detailed information about a specific optical transport interface, including its status, Tx power, and Rx power levels.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: replace optical-transport interface with , Description: Replace the degraded SFP/XFP module with a new one to restore the optical transport link.

Step 2: Step: Check and repair fiber cuts, Command: display optical-transport interface | include fault, Description: Check if there are any fiber cuts or other faults affecting the optical transport link and repair them as necessary.

Post-Check Verification

```
Command: display optical-transport interface, Expected Output:
Interface Name Status Tx Power (dBm) Rx Power (dBm), Description:
Verify that the optical transport link is operational and the SFP/XFP
module is functioning properly.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: undo replace optical-transport interface , Description: Restore the original SFP/XFP module if the replacement fails.

- Step: Restore the original fiber configuration, Command: undo display optical-transport interface , Description: Restore the original fiber configuration if the repair fails.